Capacitor: A capacitor is a device for storing charge. In actual practice, *a capacitor is an electrical device consisting of two conductors separated by an insulating or dielectric medium (including air) and carrying equal and opposite charges*. The conductors are called plates and may be of any shape.

Capacitance: Capacitance is the ability of a capacitor to store charge. *The amount of charge given to a capacitor to raise its unit potential is called the capacitance of a capacitor*. It is denoted by C and defined as,

$$C = \frac{Q}{V}$$

Where, Q is the amount of charge and V is the potential difference. The SI unit of capacitance is the *Farad (F). 1 farad is 1 coulomb per 1 volt.*

Capacitance of a parallel plate capacitor

Let us consider two similar flat non-conducting plates arranged parallel to one another, separated by distance d, as shown in fig. 1. Let A be the area of each plate and suppose that there is a charge $+Q$ on one plate and $-Q$ on the other plate. Away from the edge, the field $\vec{E}$ is very nearly uniform in the region between the plates and given by

$$E = \frac{\sigma}{\varepsilon_0}$$

$$\therefore \qquad \sigma = E\varepsilon_0 \qquad (1)$$

Where, σ is the density of the surface charge *(surface charge per unit area)* on the inner surface of the plate. If V is the potential difference between the plates then,

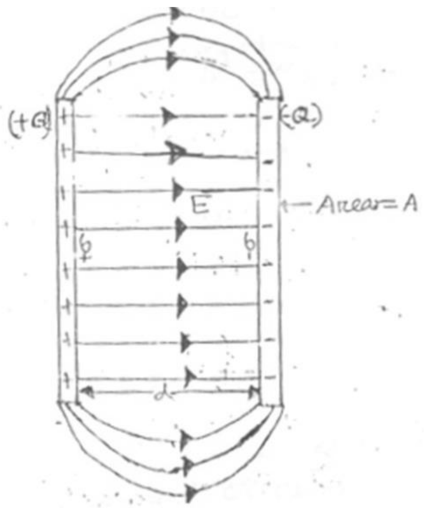

Fig. 1. A parallel plate capacitor.

$$E = \frac{V}{d}$$

Hence, equation (1) reduces

$$\sigma = \varepsilon_0 \frac{V}{d}$$

$$\therefore \qquad V = \frac{\sigma d}{\varepsilon_0} \qquad (2)$$

If A is the area of the surface then we can write

$$\sigma = \frac{Q}{A}$$

Thus, equation (2) can be written as

$$V = \frac{Qd}{A\varepsilon_0} \qquad (3)$$

And capacitance,

$$C = \frac{Q}{V}$$

$$Or \qquad C = \frac{Q}{\frac{Qd}{A\varepsilon_0}}$$

$$\therefore \qquad C = \frac{A\varepsilon_0}{d} \qquad (4)$$

If we put a medium of dielectric constant *k* in the free space between the two plates then we get

$$C = \frac{A\varepsilon_0 k}{d} \qquad (5)$$

Equations (4) and (5) are the expressions for the capacitance of a parallel plate capacitor.

From the above equations we conclude that, the capacitance of a capacitor depends on the geometrical shape of the plates of a capacitor not on *charge* and the *potential difference* between the plates of the capacitor.

Capacitance of a cylindrical capacitor

Let us consider a capacitor consisting of two coaxial cylinder of radius *a* and *b* (where *b*>>*a*) and length *l*. We construct a Gaussian surface having radius *r*, by using Gauss's law we get

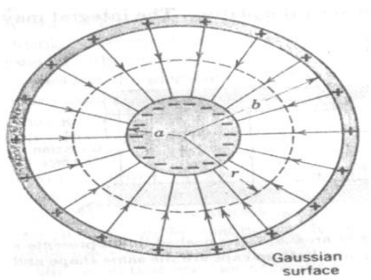

Fig. 2. A cylindrical capacitor.

$$\varepsilon_0 \oint_S \vec{E}.d\vec{a} = Q$$

Where, *Q* is the charge on the surface.

$$\varepsilon_0 E(2\pi r l) = Q$$

$$\therefore \qquad E = \frac{Q}{2\pi\varepsilon_0 r l}$$

The potential difference between the plates is given by

$$V = -\int_a^b \vec{E}\,.d\vec{r}$$

As the angle between $\vec{E}$ and $d\vec{r}$ is 180^0, since. So, $\vec{E}.d\vec{r} = -Edr$, Hence

$$V = \int_a^b E\,dr$$

$$= \int_a^b \frac{Q}{2\pi\varepsilon_0 r l} dr$$

$$= \frac{Q}{2\pi\varepsilon_0 l}\int_a^b \frac{dr}{r}$$

$$= \frac{Q}{2\pi\varepsilon_0 l}[lnr]_a^b$$

$$= \frac{Q}{2\pi\varepsilon_0 l}[ln\, b - ln\, a]$$

$$v = \frac{Q}{2\pi\varepsilon_0 l} ln\frac{b}{a}$$

Therefore, the capacitance

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{2\pi\varepsilon_0 l} ln\frac{b}{a}}$$

$$\therefore \quad C = \frac{2\pi\varepsilon_0 l}{ln\frac{b}{a}}$$

From the above equations we conclude that, the capacitance of a cylindrical capacitor is *independent of charge of the capacitor and the potential difference between the plates of the capacitor.*

Capacitance of a spherical capacitor

Let us consider a capacitor consisting of two concentric spherical shells of radius a and b, where $b>>a$. We construct a Gaussian surface having radius r, by using Gauss's law we get

$$\varepsilon_0 \oint_S \vec{E}.d\vec{a} = Q$$

Where, Q is the charge on the surface.

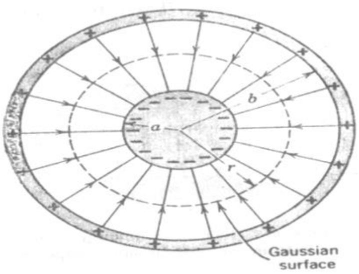

Fig. 3. A spherical capacitor.

$$\varepsilon_0 E(4\pi r^2) = Q$$

$$\therefore \qquad E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

The potential difference between the plates is given by

$$V = -\int_a^b \vec{E}.d\vec{r}$$

As the angle between $\vec{E}$ and $d\vec{r}$ is 180^0, since. So, $\vec{E}.d\vec{r} = -Edr$, Hence

$$V = \int_a^b E\, dr$$

$$V = \int_a^b \frac{Q}{4\pi\varepsilon_0 r^2} dr$$

$$V = \frac{Q}{4\pi\varepsilon_0}\int_a^b \frac{dr}{r^2}$$

$$V = \frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_a^b$$

$$V = \frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{b}+\frac{1}{a}\right]$$

$$\therefore \qquad V = \frac{Q}{4\pi\varepsilon_0}\frac{b-a}{ab}$$

Therefore, the capacitance

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{4\pi\varepsilon_0}\frac{b-a}{ab}}$$

$$\therefore \qquad C = 4\pi\varepsilon_0\frac{ab}{b-a}$$

Mathematical problems

Problem-1: *The parallel plates of an air filled capacitor are everywhere 1.0 mm apart. What must be the plate area if the capacitance is to be 1.0 farad?*

Solution: The capacitance of a parallel plate capacitor is

$$C = \frac{A\varepsilon_0}{d}$$

$$Or \qquad A = \frac{Cd}{\varepsilon_0}$$

$$Or \qquad A = \frac{1\times 0.001}{8.854\times 10^{-12}}$$

$$\therefore \qquad A = 1.1\times 10^{8}\ m^2\ (\boldsymbol{Ans})$$

Here,

$C = 1\ Farad$

$d = 1\ mm = 0.001\ m$

$\varepsilon_0 = 8.854\times 10^{-12}\ C^2/N-m^2$

$A = ?$

Problem-2: *The parallel plate of a Ruby-Mica filled capacitor are every 2.0 mm apart. What must be the plate area be if the capacitance is to be 2.0 F? Given dielectic constant for Ruby-Mica is 1.5.*

Solution: The capacitance of a parallel plate capacitor is

$$C = \frac{A\varepsilon_0 k}{d}$$

$$Or \qquad A = \frac{Cd}{\varepsilon_0 k}$$

$$Or \qquad A = \frac{1\times 0.002}{8.854\times 10^{-12}\times 1.5}$$

$$\therefore \qquad A = 1.5\times 10^{8}\ m^2\ (\boldsymbol{Ans})$$

Here,

$C = 2\ Farad$

$d = 2\ mm = 0.002\ m$

$\varepsilon_0 = 8.854\times 10^{-12}\ C^2/N-m^2$

$k = 1.5$

$A = ?$

Problem-3: *A parallel plate capacitor has a capacitance of 100 μF, a plate area of 100 cm^2, and a mica dielectric constant ($k = 5.4$). If the potential difference in between the parallel plate is 50 volt, calculate*

(i) Electric field in the mica

(ii) The free charge on the plate and induced charge.

Solution: Free charge is

$$q = C_0 V_0$$

or $$q = 100 \times 10^{-6} \times 50$$

∴ $$q = 5 \times 10^{-3}\ C(\boldsymbol{Ans})$$

Electric field in the mica

$$E = \frac{q}{A\varepsilon_0 k}$$

or $$E = \frac{5 \times 10^{-3}}{1 \times 10^{-2} \times 8.854 \times 10^{-12} \times 5.4}$$

∴ $$E = 1.04 \times 10^{10}\ volt\ (\boldsymbol{Ans})$$

Induced capacitance is $C_{ind} = C_0 k = 100 \times 10^{-6} \times 5.4$

$$C_{ind} = 5.4 \times 10^{-4}\ F$$

Induced charge is

$$q_{ind} = C_{ind} V_0$$

or $$q_{ind} = 5.4 \times 10^{-4} \times 50$$

∴ $$q_{ind} = 2.7 \times 10^{-4}\ C(\boldsymbol{Ans})$$

Here,

$C_0 = 100\ \mu F = 100 \times 10^{-6} F$

$A = 100\ cm^2 = 1 \times 10^{-2}\ m^2$

$\varepsilon_0 = 8.854 \times 10^{-12}\ C^2/N - m^2$

$k = 5.4$

$V_0 = 50\ volt$